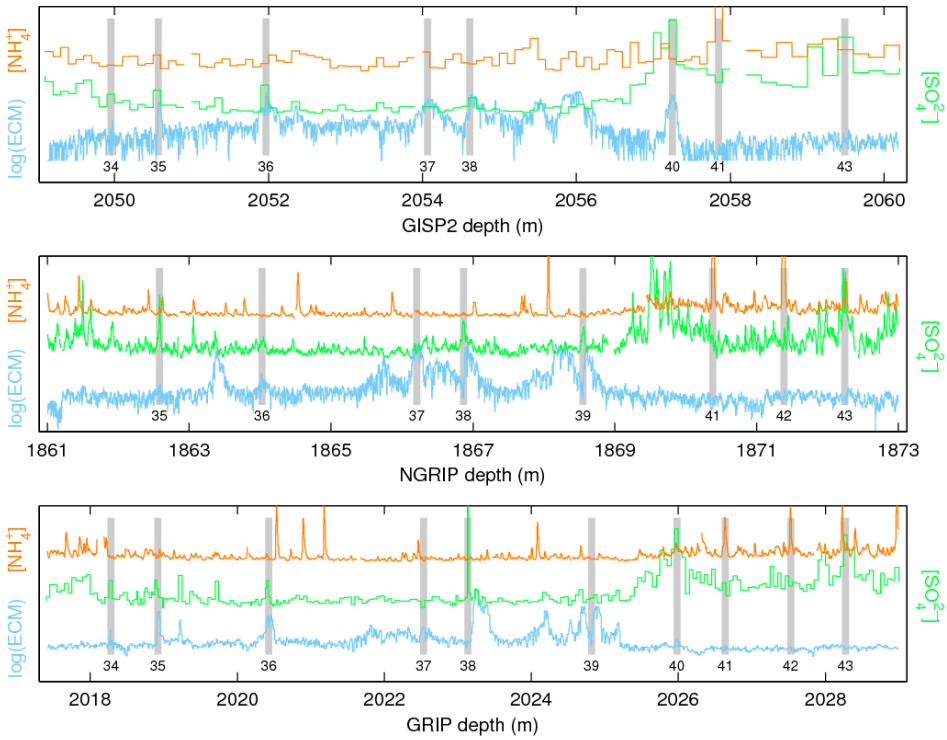
Matchmaker

A MATLAB tool for manual matching of (paleoclimate) time series

User's guide



Matchmaker was originally constructed by Sune Olander Rasmussen.
This is the user's guide for the revised version by Sune Olander Rasmussen and Giulia Sinnl.

Report GitHub issues: https://github.com/giulia-s/matchmaker/issues
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Introduction

Matchmaker is a collection of MATLAB scripts that form an application aiming at facilitating efficient and quality-checked matching of time series from different records. It is also convenient for browsing and inspecting data sets with many parallel data series from one (or up to a couple of) records.

It has been designed for use with ice core profiles, and some design choices clearly reflect this, but can essentially be used with any type of data, although the tool is probably not suited for working with data sets from many independent cores/sites due to the screen layout.

This User's guide is meant to provide the user with all relevant instructions on how to operate matchmaker, assuming no previous knowledge. *It is recommended to read all parts carefully before starting to use the program.*

Copyright

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When using this script please cite:
Rasmussen, S.O., Seierstad, I., Andersen, K.K., Bigler, M., Dahl-Jensen, D., and Johnsen, S.: Synchronization of the NGRIP, GRIP, and GISP2 ice cores across MIS 2 and palaeoclimatic implications, Quaternary Sci. Rev., 27, 18–28, https://doi.org/10.1016/j.quascirev.2007.01.016, 2008.

You are free to edit the source code to fix malfunctions, add features, or change details in layout or user interface, but if you think that your changes represent an improvement to all (or even most) users, forward your suggestion to Sune for inclusion in future versions instead of changing your local version.

Matchmaker is not foolproof (and will never be) and you may still be able to provoke errors that will cause loss of data. Therefore, please act responsibly and back up your data and match point files frequently.

Technical issues and preparation

MATLAB version

The revised version of Matchmaker has been tested on MATLAB R2020b and R2024b. Older version exist but are not on GitHub.

Platform

Matchmaker has been developed on Windows XP/2000 Pro/7 platforms but is known to work under Linux as well.

Files and folders

The program can run from any folder.

The program consists of the following files:

1) *matchmaker.m*
2) *matchmaker_evaluate.m*
3) *matchmaker_savefig.m*
4) *matchmaker_sett.mat*
5) *stepit.m*
6) *files_main.mat (user defined)*
7) The mandatory subfolder *data*, containing the appropriate data files for each ice core, as described on page 4.
8) The mandatory subfolder *matchfiles*, containing the appropriate data files for the tie points saved during a matchmaker session. Also, automatically generated by the program: the *matchfiles/matchfiles_backup* folder, containing backup versions of the matchfiles, as described on page 5.
9) A template file for building Matchmaker data files *make_data_file.m* is also provided.
10) This user's guide.

Format of input data

Files_main: initializing the project files

You need to supply Matchmaker with a list of which ice cores and files to work with, all stored in a struct called *files*. Use the file "files_main.m" as an example to set the initializing files of the project. Be aware of the numbering used. For example:

files.core{1} = 'Core1';
files.datafile{1} = 'core1_data.mat';
files.matchfile{1} = 'CORE1_core123match.mat';
files.matchfile_bis{1} = 'CORE1_core123match_version2.mat'; %(optional)

NOTE: Naming convention:
It is important to note that, although one match-file is needed for each data-file, the match-files of a certain matching project belong together.

To prevent mistakes, it is recommended to use a naming convention where both the ice-core name and a suitable name of the matching project identify the match-files. For example, in a project where three cores are matched, one could for example call the initializing file ‘core123match_main.m’ and the match-files 'CORE1_ core123match.mat', ‘'CORE2_ core123match .mat', and "CORE3_ core123match .mat'.

Be careful to keep track of which files are used: if 3 records are matched up, nothing prevents you from opening Matchmaker with only the first two records and remove or add match points (mp). However, if you modify the number of match points, the third record will no longer be matched to the two others, because Matchmaker assumes that all three match point files contain parallel ordered lists of match points that are valid in all three cores. To solve this problem, make sure to take advantage of the ‘dummy’ horizons, as explained in the section “Types of match points”.

Data format (files.datafile)

An example of a MATLAB script for building Matchmaker data-files is provided (*make_data_file.m*). The format is designed to handle that most ice core data are available on common depth/time scales, and that it is often unnecessary to save a depth/time scale for each data series.

In short, the data file must be an internal MATLAB data format (.mat) file containing the following variables:

data cell array with N data series.

depth cell array with M depth (or time) scales.

depth_no vector of size N. Element *i* indicates which element *j* of *depth* belongs to the *data* series *i* (see ***Figure** 1*).

species cell array with the names of the data series, of size N.

colours Nx3 array with RGB colour definition used for the plotting.

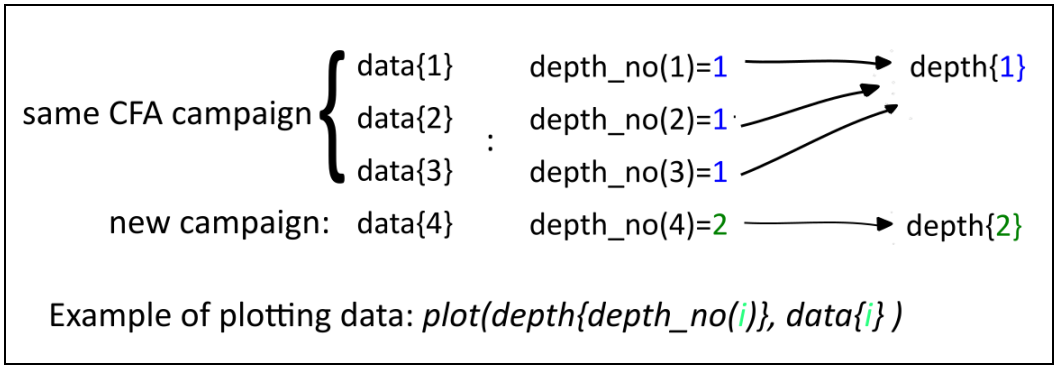


Figure 1 Schematic of how the arrays data *and* depth *are connected by* depth_no.

Match-point file format (files.matchfile and files.matchfile_bis)

Match points are saved in the *matchfiles* folder, with one file for each ice core. Each match-file must contain the array *mp* with 2 columns: depth (or time) and match-point type (see: "Types of match points). This array is populated during a matchmaker session and then saved by the user.

If *files.matchfile(i)* is not found upon opening the program, Matchmaker will create a new file named *files.core(i).mat*. The program will check for the existence of this file, and either load it or initialize an empty array *mp=[]*. You can keep working with this file but remember to eventually set a naming convention suited for your project.

An optional, secondary file *matchfile_bis* can be added for each ice core to be able to compare a secondary set of tie points. See page 15 for how to use this file.

Using Matchmaker

Starting Matchmaker

The user can freely select how many ice cores and how many data series to show for each core.

Matchmaker is started by issuing the command in the MATLAB prompt, for example:

```
>> matchmaker('files_main', [1 2 3], [2 3 4])
```

In this example, matchmaker reads from the initialization file *files_main* and opens the cores with number 1, 2, and 3. Each ice-core panel opens with 2, 3, and 4 data series, respectively.

When the Matchmaker window has successfully opened, you start by choosing which of the data series to view and which depth/time interval to view them in (follow section on Scaling the axes). You are now ready to look for common features in the different cores, match the records using mouse clicks, and visualize the results by stretching the data ("accordianizing") or by using the basic evaluation tools provided.

Errors

Errors occur, although care has been taken to eliminate critical errors. MATLAB warnings also occur on a regular basis.

In case of critical errors or warnings that sound worrying, cut-and-paste the error message from the MATLAB prompt and send it to me by e-mail with a description of the action that caused the error to happen (or post it as an issue on the GitHub of matchmaker).

Most errors are produced within a callback, and often the application will be able to continue without problems. However, it is probably smart to save the match points and restart Matchmaker.

Closing down after critical errors

Following a full-fledged crash, it can be impossible to get MATLAB to close the windows of Matchmaker or the evaluate window. The brute-force close-down procedure is this:

```
set(0, 'showhiddenhandles', 'on')
delete(gcf)
set(0, 'showhiddenhandles', 'off')
```

This series of commands closes the most recently accessed figure without any warnings or undo options. If you are trying to kill e.g. the *Evaluate* window, be careful not to kill the Matchmaker window (without saving) by accident. It is *not* a pretty solution, but better than closing MATLAB or using Ctrl+Alt+Del.

Types of match points

Seven types of match points can be set in the standard version of Matchmaker (***Figure 2***), and here we list the common convention for using them.

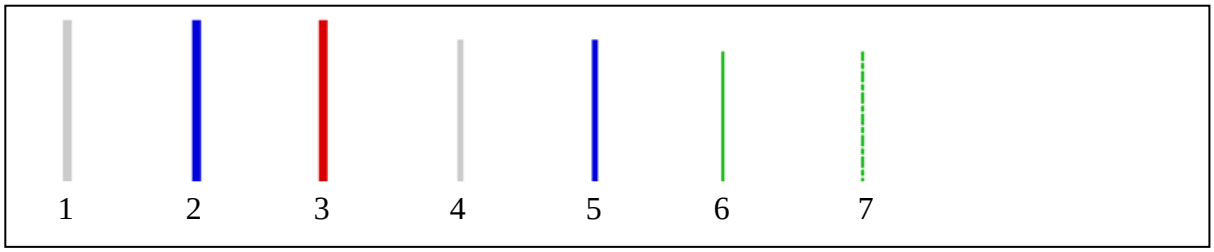


Figure 2 Types of match points in Matchmaker

To set new match points, first read the paragraph “Edit” in the section “Buttons at the bottom of the screen”.

Here, we recommend a standard notation on how to assign meaning to the match point types. There are three main categories of match points:

- **Reference match points**, also known as the main isochrones or reference horizons, used to match ice cores together. They represent simultaneous event, and when pressing the *accordianize* button, the data series are stretched horizontally so that the first and last reference match points on the screen line up vertically. The reference match points come in three varieties:
  - **Thick grey bars, numbered** (in the code: mp type 1), set at depths where common features are observed in the different records, such as volcanic eruptions.
  - **Thick blue bars, numbered** (mp type 4), behave similar to grey bars but represent a "dummy" match point used when a feature is observed in some, but not all records. In that case, a dummy match point is inserted in order to maintain correct numbering of the match points. They are for example used when three records are matched and a certain feature is only seen in two records. The feature is marked by a grey marker (mp type 1) in the two records but has to be added to the third record as a blue marker (type 4) to maintain the same overall number of match points. Type 4 match points are numbered and are equivalent to type 1 match points when accordianizing, but are represented with a different symbol when *evaluating*.
  - **Thick red bars, numbered** (mp type 3), are intended as “temporary” versions of type 1 match points. Often the matching is initially performed on long data sequences inspected in poor detail. Type 3 match points are conventionally set to mark likely match points, and a second inspection using high resolution data is then undertaken to replace the temporary match point with a type 1 match point (or to delete the match

point). Type 3 match points are equivalent to type 1 match points when accordianizing or evaluating.

- **Thin mps**, are minor mps used to mark features in the data which support the matching but are not match points in themselves:
  - **Thin grey bars, not numbered** (mp type 2), are match points representing finer, less certain features, and are usually used to mark features or patterns that are likely to be synchronous but not in themselves convincing enough to be match points used for the synchronization. They play no role for the stretching of the data series.
  - **Thin blue bars, not numbered** (mp type 5), are “dummies” for type 2 match points, just as type 4 match points are for type 1 match points.
- **Years**, the minor mps used to mark the annual layers in each ice core. They come in two types:
  - **Thin green bars, numbered** (mp type 6), are used to mark annual layers. The numbering follows these principles: if there are no reference match points on screen, the numbering starts at the left-most edge of the screen; if there are reference match points on the screen, the numbering starts at the first layer after (i.e., to the right of) the left-most reference match point.
  - **Dashed thin green bars, numbered** (mp type 7), are used to mark annual layers that are *uncertain*. They behave as type 6 markers in terms of numbering.

Too many match points

When too many match points are within the window displayed, Matchmaker goes into ”space-saving mode” and will not plot all year markers and minor matchpoints. Instead, a warning is shown, encouraging the user to reduce the x-axis range. The accordianizing still works as usual, but you can only modify the types of match points which are shown.

Buttons at the bottom of the screen

Figure 3: Button row of Matchmaker (located below the data axes), cut in two.

The ***Save*** button saves the match points (surprise, surprise) by overwriting all the matchfiles. Before saving, the previous version of each of the match point files is copied to the same name with a ".backup" extension. If Matchmaker crashes while saving, you should be able to restore the previous version by simply renaming this file. Additionally, a backup folder is created programmatically to store the run: *matchfile*, *matchfile_bis* (if present), and *files_main*. The backup folders are named *matchfiles/matchfiles_backup/ files_main /#N*, where *#N* is a counter increased every time the 'Save' button is pressed.

The following saving strategy is recommended: whenever an important version is saved, the corresponding *#N* folder should be renamed to something relevant to the user (e.g. *#N_revised_mps_due_to_new_data*). In this way one can delete unnecessary backup folders and quickly find the most important version. The counter does not care if folders are removed or renamed, it will independently continue increasing upon pressing *Save*.

Note: To load a backed up version, simply paste the content of the relevant backup folder into the *matchfiles* folder.

The ***SaveFig*** button creates a simplified, editable screenshot of the user interface. The external function *matchmaker_savefig* takes care of this process. The user is prompted to insert a figure title in the command line:

>>> Insert title [with apex!!]: 'new_figure'

The name should be enclosed in quotation marks, without any file-name extension (i.e. no '.fig' needed).

A figure called *new_figure_date.fig* will be produced. One can add a timestamp too, if desired, by uncommenting the line where *title_str* is defined.

The ***Accordianize*** button stretches the data series horizontally to align the reference mps (type 1/3/4). The number specified in the small text field next to the button is the *leading data panel* (counting from the bottom up). The accordianization leaves the scaling of the leading data panel unchanged and uses the reference mps in the leading panel to scale the remaining data panels according to the following rules:

- If there are two or more reference match points visible in the leading data panel, the remaining axes are scaled so that the first and last of the type reference horizons are aligned.
- If there is only one reference match point visible in the leading data panel, the other axes are scaled so that the corresponding match point is aligned between the cores, and so that the interval shown has the same length as in the leading data panel.
- If no reference match points are visible, nothing happens.

The ***Evaluate*** button opens another window, if at least two match points are visible in the leading data panel, and these match points have been set in the other records as well. The window can be left open or closed after use. If left open, a second press on the *Evaluate* button in the Matchmaker main window updates the match-point data and brings the *evaluate window* to the front without changing the axis scaling.

Description of the Evaluate window On the left graph, the depth differences between match points are plotted, versus the depths of the match points in the leading record. The depth differences are normalized to zero-mean for the section shown. This plot can be replaced by a depth-versus-depth plot by toggling the *Depth diff* button. In both cases, the filled dots are type 1/3 match points, and the open dots are type 4 match points. Minor mps (type 2/5 & 6/7) are not evaluated.

The right plot shows the relative difference between neighbouring match points (i.e., the slope of the matching curve). Inaccurate or erroneous match points can often be spotted in this panel as an interval with anomalous slope.

When clicking on a match point on the left graph, a dashed line is plotted in the corresponding position on the right graph, and the match point number is shown in the *Match point ID* text field at the bottom.

The x-axis scaling (i.e. the match point interval) can be changed using the text fields below the graphs. The small middle field is the step size used when browsing through the match points using the keyboard arrow keys.

Zooming can be done using the built-in MATLAB zoom-function in the menu bar. If you want to return to automated scaling, right-click on the axes and choose "reset to original view". Alternatively, close and re-open the evaluate window from Matchmaker.

NOTE: If the *Show secondary* button is activated in Matchmaker, the secondary set of tie points will be shown as an additional line on the evaluation plots.

The ***Edit*** button controls whether match points can be set or erased by clicking the axes. Once activated, the match-point categories are selected from the “radio buttons” to the right:

- Reference mps: edit mp of type 1 (left-click), 4 (right-click), and 3 (shift-click)
- Thin mps: edit mps of type 2 (left-click) and 5 (right-click)
- Years: edit mps of type 6 (left-click) and 7 (right-click)

The ***Hide minor mps*** button controls whether minor mps (thin and years) are shown on screen. The minor mps can only be added or removed when shown, i.e. when the button is toggled off.

The ***Show secondary*** button is used when comparing two sets of match points (see the dedicated section “Comparing two sets of match points).

The ***Check mps*** button counts the number of match points and presents deviations between records in a table at the Matlab prompt. The smallest distance between neighbouring match points is also shown, as very proximal thin and annual markers can be very hard to distinguish on the screen.

The ***Undo*** button reverses any addition/removal of tie points, whatever their type. There is no “redo” function, i.e. each undo is permanent. The program saves 1000 moves, but this can be changed in handles.N_undo. Once this value is reached, the first moves will start to be deleted from memory.

The ***Exit*** button, Windows' x in the upper right corner, and *Alt-F4* closes Matchmaker. If you are trying to exit without saving changes to the match points, you are asked twice to confirm.

Keyboard commands

Press space bar (or any undefined key) to see a list of keyboard shortcuts (*Figure 4*)

Pressing the arrow keys will move the x-axis of the leading data panel right or left by the distance given below the arrow buttons in the leading data panel. Subsequently, the other axes will be accordianized. This feature is particularly useful when browsing through and inspecting a set of existing match points.

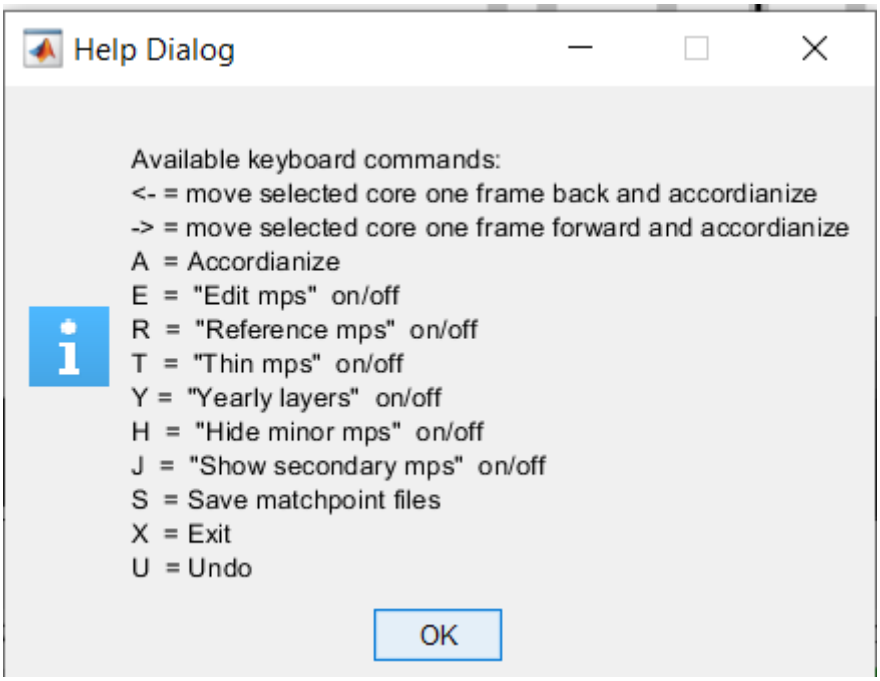


Figure 4: Keyboard shortcuts in matchmaker

Scaling the axes

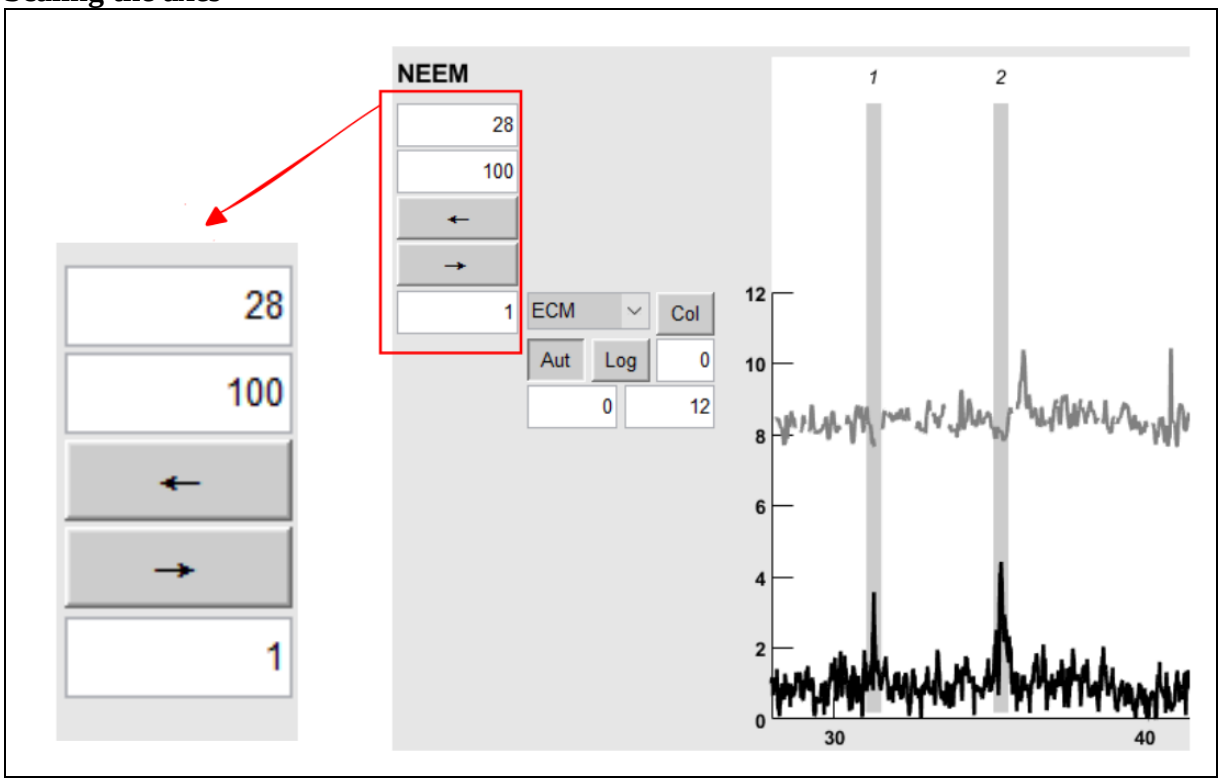


Figure 5: Depth-axis buttons

Buttons for scaling of the depth axes are located to the left of each data panel (***Figure 5***). The field below the arrow buttons contains the increment used when using the arrows (both the keyboard arrows or the buttons on screen) to move the axes.

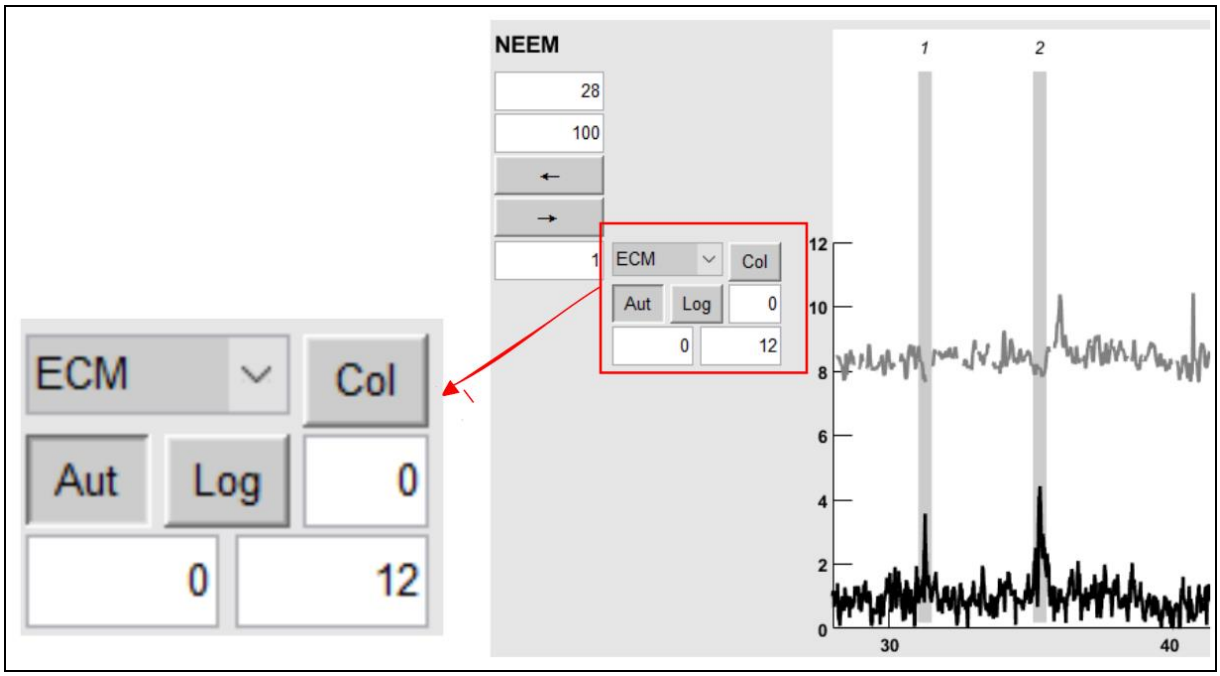


Figure 6: Species' button group

Each data sub-panel has a cluster of buttons (*Error! Reference source not found.*).

The drop down list enables the user to change which data series (*species*) is shown. Next, the *Color* button can change the color of the data line, the new colour is automatically updated in the data file.

The *Aut* button re-enables automatic scaling after manual scaling has been applied, and the *Log* button toggles logarithmic y-axis on/off.

The field next to *Log* moves the species horizontally relative to the depth axis by the specified amount. This feature is used to correct for an incorrect depth registration of other problems, but can easily lead to confusion. Note that the match points are still placed relative to the depth axis shown, not relative to the offset data series. Use is generally not encouraged, and the field lights up in red when a non-zero value is entered.

The two fields underneath set the y-limits of the species.

Comparing two sets of match points

If you want to compare or review two parallel sets of match points **on the same ice core,** you can either open Matchmaker twice (potentially from different libraries) with the different files and settings, or instead display and edit the two sets of match points on one screen.

To do this, for the ice core in question, add a secondary set of mp under the matchfile_bis variable in the files_main file. E.g.:

files.core{1} = 'Core1';

files.datafile{1} = 'core1_data.mat';

files.matchfile{1} = 'CORE1_core123match.mat';

files.matchfile_bis{1} = 'CORE1_core123match_version2.mat';

Then open Matchmaker and press the ***Show secondary*** button. The primary match points will be reduced in height and the secondary set will appear in the lower half of each core window.

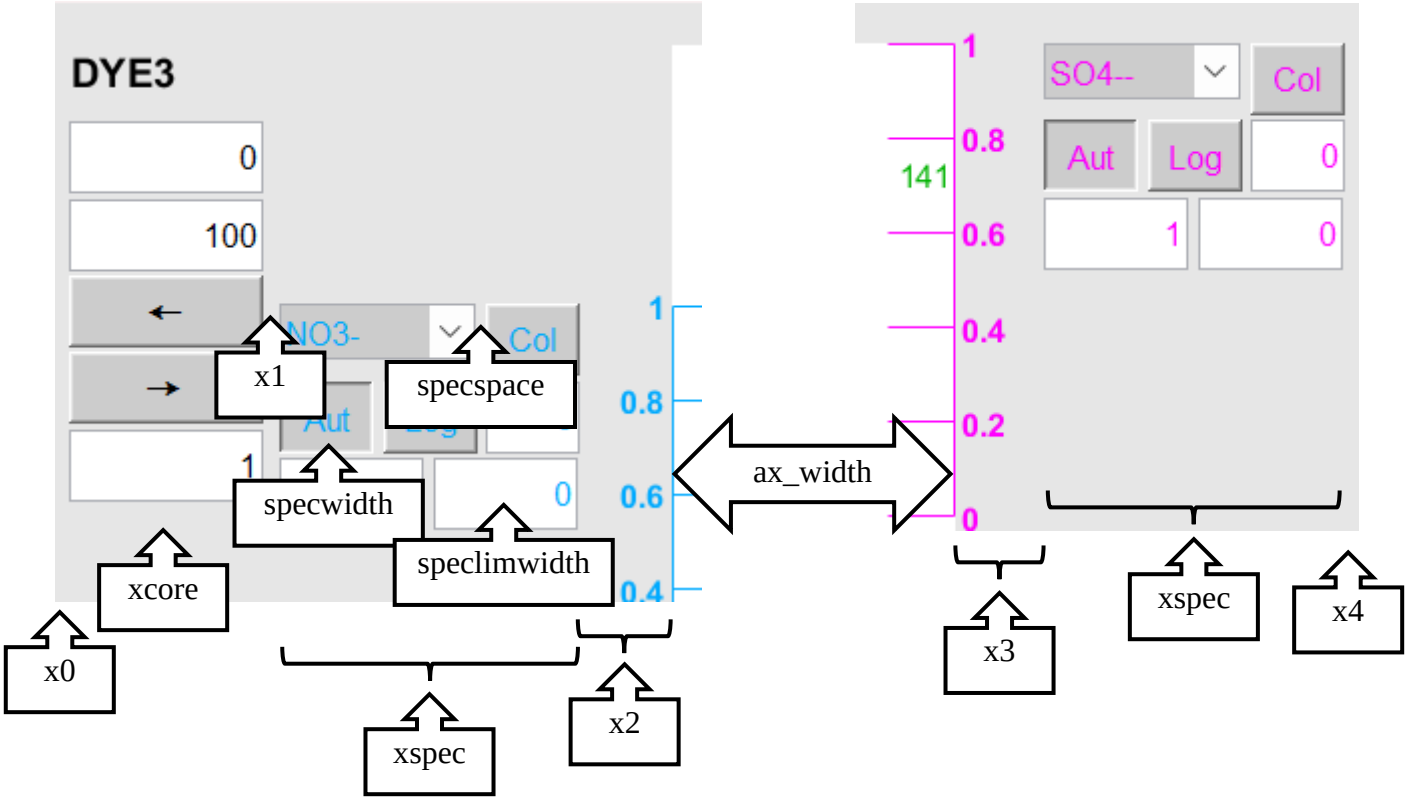
Match points are modified as usual by pressing ***Edit*** and the preferred type.

Upon pressing ***Save*** all files will be saved and the backup will be produced as usual.

Accordianizing only uses the top match points.

Evaluate will display the secondary set as a dashed line.

Appendix: Names of the distances used for drawing the Matchmaker user interface



Note that

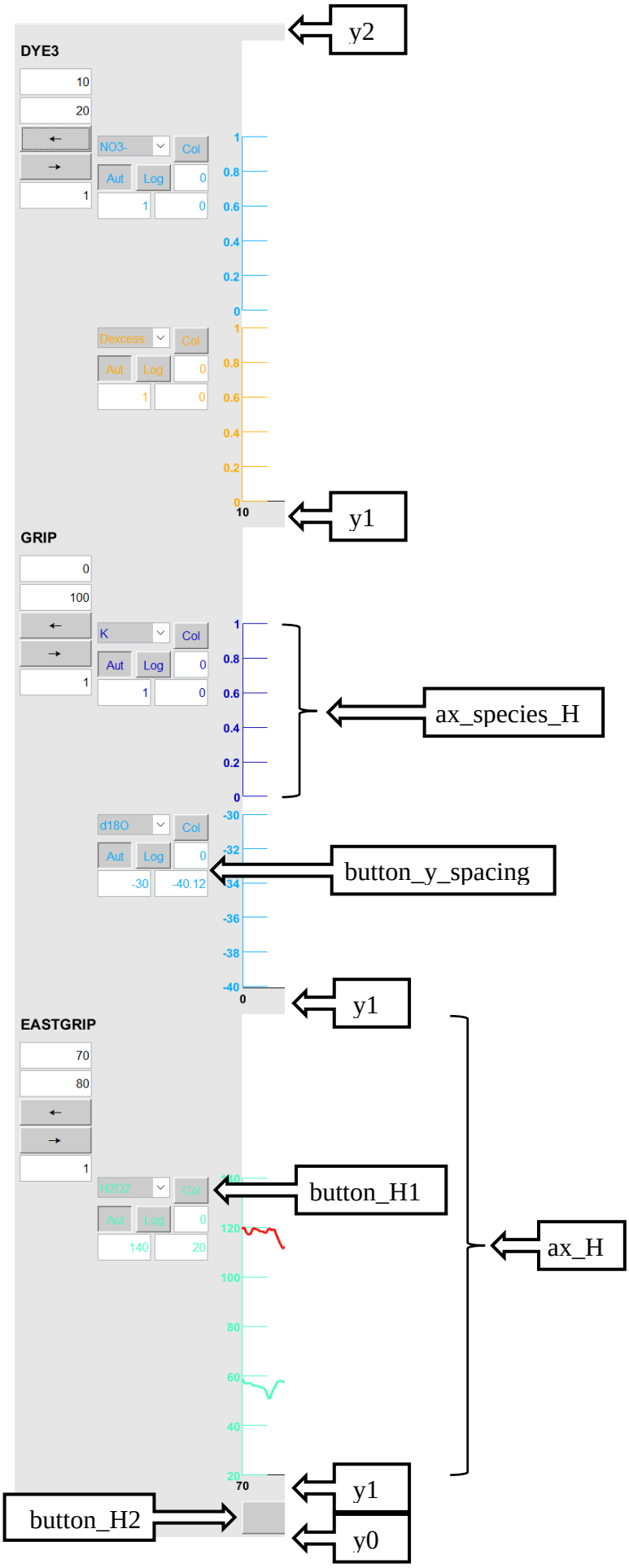
- "specspace" (horizontal button separation in the species block),
- "specwidth" (width of the Col, Aut, and Log buttons), and
- "speclimwidth" (width of axis limit fields)

are all given as a fraction of "xspec".

Other distances:

button_x_spacing is the horizontal button separation used for the bottom buttons.

button_L is the button width for the large buttons calculated by sharing the available space below the axes. If buttons get too narrow, more space is available by allowing the buttons to expand to either side of the data axes.


y2
DYE3
10
20
NO3-
Col
Aut
Log
0
1
0
Dexcess
Col
Aut
Log
0
1
0
10
y1
GRIP
0
100
K
Col
Aut
Log
0
1
0
ax_species_H
d18O
Col
Aut
Log
0
-30
-40.12
button_y_spacing
0
y1
EASTGRIP
70
80
H2O2
Col
Aut
Log
0
140
20
button_H1
ax_H
70
y1
button_H2
y0